

Master in Artificial Intelligence and Robotics
Elettive in Artificial Intelligence: NUANS
Academic Year: 2022-23

NetSheeran: Lyric Remixer with Summary Title



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Goal

Construct an architecture capable of remixing song verses and arranging them into a new song.

Apply an abstract summarization pipeline to this newly generated song to determine a short title.

Dataset

1. MoodyLyrics

- a. <http://softeng.polito.it/erion/>
- b. developed by Erion Çano lab (Politecnico di Torino) referring to another study
<https://cs.nju.edu.cn/sufeng/data/musicmood.htm>

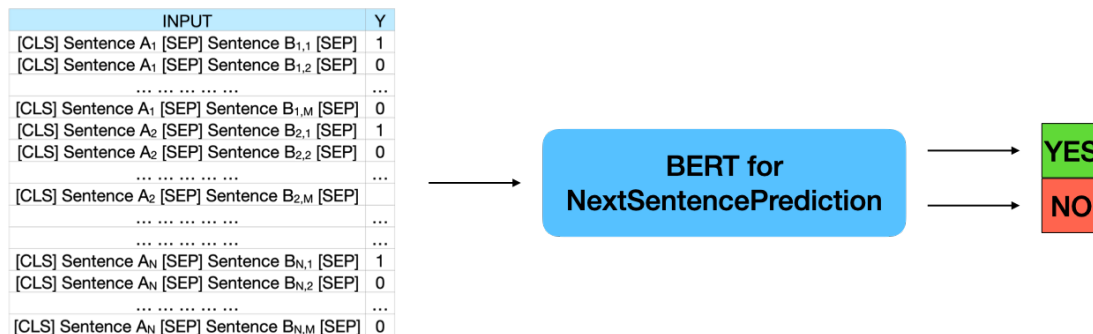
2. Genius (lyrics provider)

- a. accessible through Genius Api.
- b. to retrieve songs' lyric.

The Approach

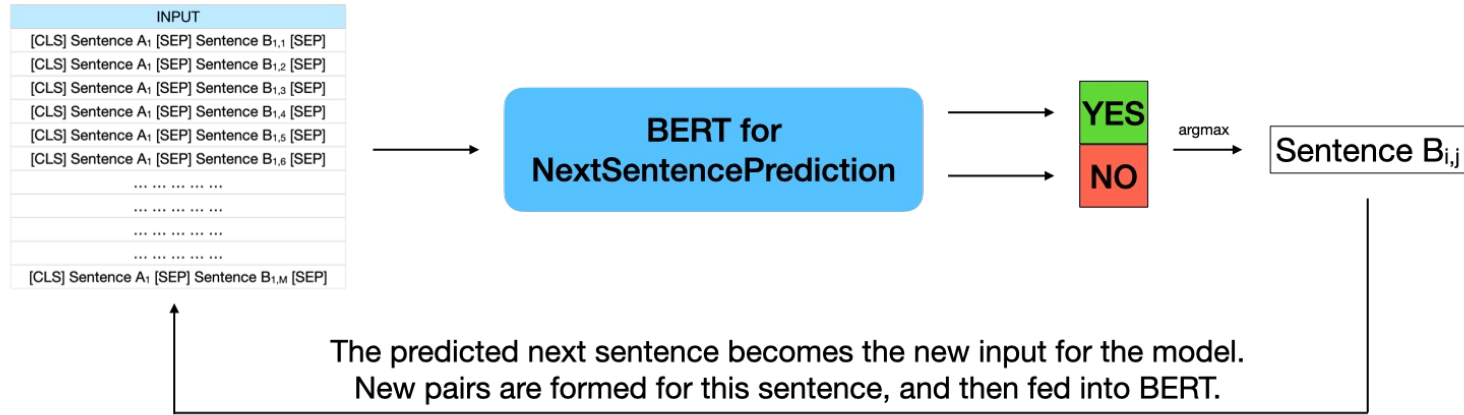
Phase	Task	Architectures
First phase: Learn to mix verses from existing lyrics	Fine-tuning BERT to fit the prediction of the next sentence to our task domain.	BERT for NextSentencePrediction
Second phase: Remix verses to create a new song lyric	The new song lyric is generated by sampling verses from a test set, and argmax is used to select the best possible successor, which is determined by BERT prediction.	BERT for NextSentencePrediction
Third Phase: Abstractive Summarization pipeline for Title Generation	The T5 model generates a title from a short summary, while BERT Base measures similarity to the entire new song text.	HuggingFace Sumamarization Pipeline(t5-large) + BERT Base

Training Architecture



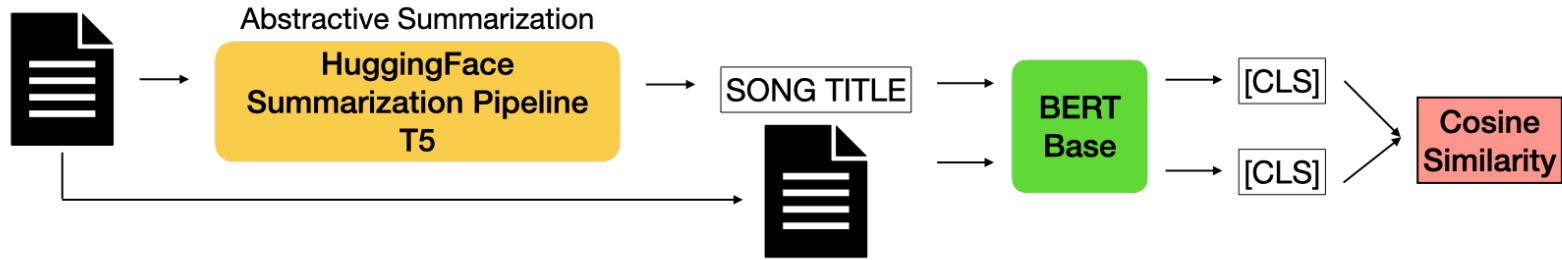
1. Loss is backpropagated through the model to fine tune the BERT architecture for the task domain.
2. Accuracy is computed in terms of BinaryAccuracy between the model's prediction for the two matched sentences and their ground truth.

Text Creation



Sentence A₁ is the first prompt (which is then iteratively replaced by the predicted Sentence B_{i,j}). The result of this execution pipeline is a new lyric of a given length, which then feeds into the summarization pipeline for the title assignment.

Title Assignment



The **cosine similarity** between BERT Base [CLS] Embedding of the title and the song is used to give an indication, in terms of Coherence, of how much the title captures the meaning of the entire text.

Final Result

1. Specifying the length of the final song text in terms of verses, the model can generate a new text by sampling verses from a test set and determining when they match.
2. The new text passes through the summarization pipeline, which assigns it a title based on the short summary.
3. For completeness, the cos-similarity between the [CLS] song embedding and the [CLS] title embedding is computed to give an indication of how coherently the title captures the new song meaning.

Thank you for your attention